

COOKING SCHOLAR

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9 - CUSTODIO, ZOOEY CARLISLE

11- DE LA PENA, DON EMMANUEL

13 - ESPIRITU, MARRIENE ELLAISA

18 - ILAO, MC BRAYDEN

CHAPTER I - PROBLEM IDENTIFICATION AND PROPOSAL

INTRODUCTION

In today's fast-paced life, wherein time is very scarce and valuable, the ability to search for the perfect dish and recipe has transformed from a necessity to a luxury. Most of the time, people come home exhausted and unable to think thoroughly about what to cook, so they order food from delivery apps or simply eat outside.

However, these habits can have long-lasting effects, too much fast food can pose a serious risk to your health. Eating outside can also be expensive and troublesome. Even with search engines and the internet, it still takes time to find a recipe; most of the time, users do not have the required ingredients. Everyone cannot simply buy ingredients for one recipe; they are most likely to buy raw ingredients and think of whatever is possible with the resources.

Knowing the right recipes can save you time and money and provide great nutritional value. According to studies, cooking your own meals can positively impact your mental health and overall psychological well-being (Farmer & Cotter, 2021). With the right tools, anyone can always cook for themselves and others.

In simple terms, not everyone cooks for themselves anymore; they usually order and eat outside. Everyone should be able to cook their own meals while saving time and resources. This also results in a healthier lifestyle, good mental health, and many more benefits.

OBJECTIVES

Cooking Scholar will provide aid for everyone who struggles with cooking and does not have the time to think about a recipe. This cuts the time and energy needed for searching for a recipe that is available and also healthy. The program aims to simplify their work and to provide more time for them. Recipes in the program should be healthy and insightful for the users to continue their day-to-day lives.

Searching is easier, and at the same time, the outputs are high quality. It also hopes to solve problems regarding exhaustion. Lastly, the user interface is friendly and easy to use because this is to minimize the stress and thinking power while using it.

PROBLEM DECOMPOSITION

- **Time Scarcity and Mental Fatigue:** As mentioned in the above paragraphs, people often eat take-out or order online for the following reasons: lack of energy, tight schedules, or lack of available resources. This habit can negatively affect health, finances, and overall well-being.
- **Limited Resources:** These days, people are too exhausted to cook after returning from work/school– let alone think about what to cook. Searching for recipes online and following them is time-consuming, which would lead to choosing the quickest choice– fast food. Online recipes tend to contain extremely niche and specific ingredients and/or tools that are not always available at home or nearby stores. Buying these ingredients and/or tools for a one-time use is irrational and expensive, especially considering today's economy.
- **Over-reliance on Convenience:** Although fast food deliveries are convenient and easy, they could pose significant health risks in the long term. Eating out is also quite expensive.
- **Health Concerns:** Relying too much on convenience increases the risks of lifestyle-related diseases such as obesity, diabetes, heart disease, high blood pressure, and fatty liver diseases because of its processed ingredients and fatty sugars. The lack of home-cooked meals reduces control over nutrition. Skipping cooking leads to missed opportunities for eating healthy. Cooking at home has proven therapeutic for mental health and relationships (Farmer & Cotter, 2021).

PROPOSED SOLUTION

Through Cooking Scholar, users can find the ideal recipe for their situation. The mobile app features a UI interface with buttons where they are to input the ingredients available for them, the equipment they have, the time they are going to allot, and finally, the output where the user can see the recipe options for them. There are also some buttons that the user selects to tailor to their needs.

- Log-in page - allows the user to log in or sign up for a new account
- Tutorial page - welcomes and guides the user on how to use the app properly
- Main page - page that shows daily and recommended recipes, and will show the icons for text box, search bar, and saved recipes
- Text box - the user will input the ingredients they have, how much time they'll need, and the equipment available (each inputs have their own text box)
- Allergen button - will enable the option to add some allergens to filter recipes
- Food database - where all information about foods reside, data here will be used and based on the user's input
- Food pages - database would use data based on the user's input, then show some preferred pages, and would show instructions, ingredients needed and cooking time
- Download button - will show every time the user is in a page, the downloaded page will be saved into the saved recipes
- Saved recipes - downloaded recipes show here, makes the recipe available offline
- Search bar - user can search for more recipes based on what they want

SCOPE

Cooking Scholar will aid students who require quick and nutritious meals. The platform for it will be a mobile application that needs the internet. Its core features are:

- The ability to input available ingredients, cooking duration, and equipment.
- Generate recipe suggestions based on gathered data.
- Save or favorite recipes for offline use.
- Add allergens to filter recipes.
- Search for and learn new recipes.

Some of its limitations include:

- Does not provide grocery delivery or shopping services.
- Does not replace professional dieticians or nutritionists.
- Requires an internet connection for full app accessibility.

Its purpose is to help users save time and money, to promote a healthy diet and lifestyle, and to provide convenience to students and learning opportunities in cooking.

CHAPTER 2 - METHODOLOGY AND DEVELOPMENT

Strategy of Development

The problem had already been decomposed into multiple features. This made it easy to modularize the project. The developers split up the features into different modules. These modules are mini-programs that perform a specific task, created by developers. A Google Drive was created for developers so that their work can be stored and accessed by other developers.

Although there was no agreed-upon naming convention, the developers had self-explanatory variable names so the others could understand them. After all modules are finished, the compiler downloads the programs. Using object-oriented programming, the files are imported into one another in the form of functions.

The project is made entirely out of Python due to its flexibility and ease of use for student programmers. A file named 'main.py' is tasked to utilize and run the functions. Then, these functions have a return value telling 'main.py' what to perform next. The program only has a text-based User-Interface (UI)

Current State of The Project

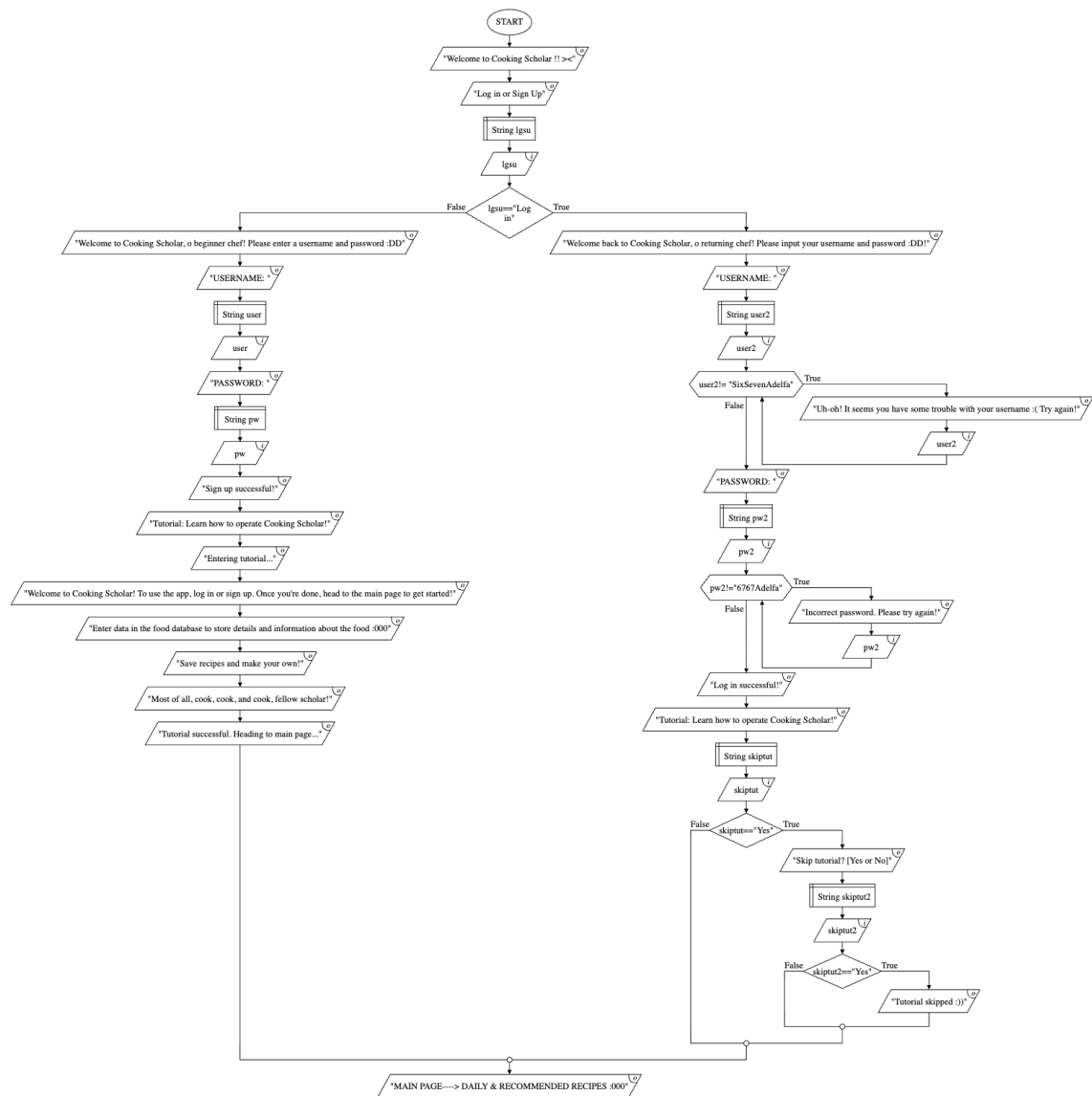
There are multiple versions of the program in the form of folders (v1, v2, etc.) This is to help distinguish between the additions, changes, deletions, and bugs that are present in each version of Cooking Scholar.

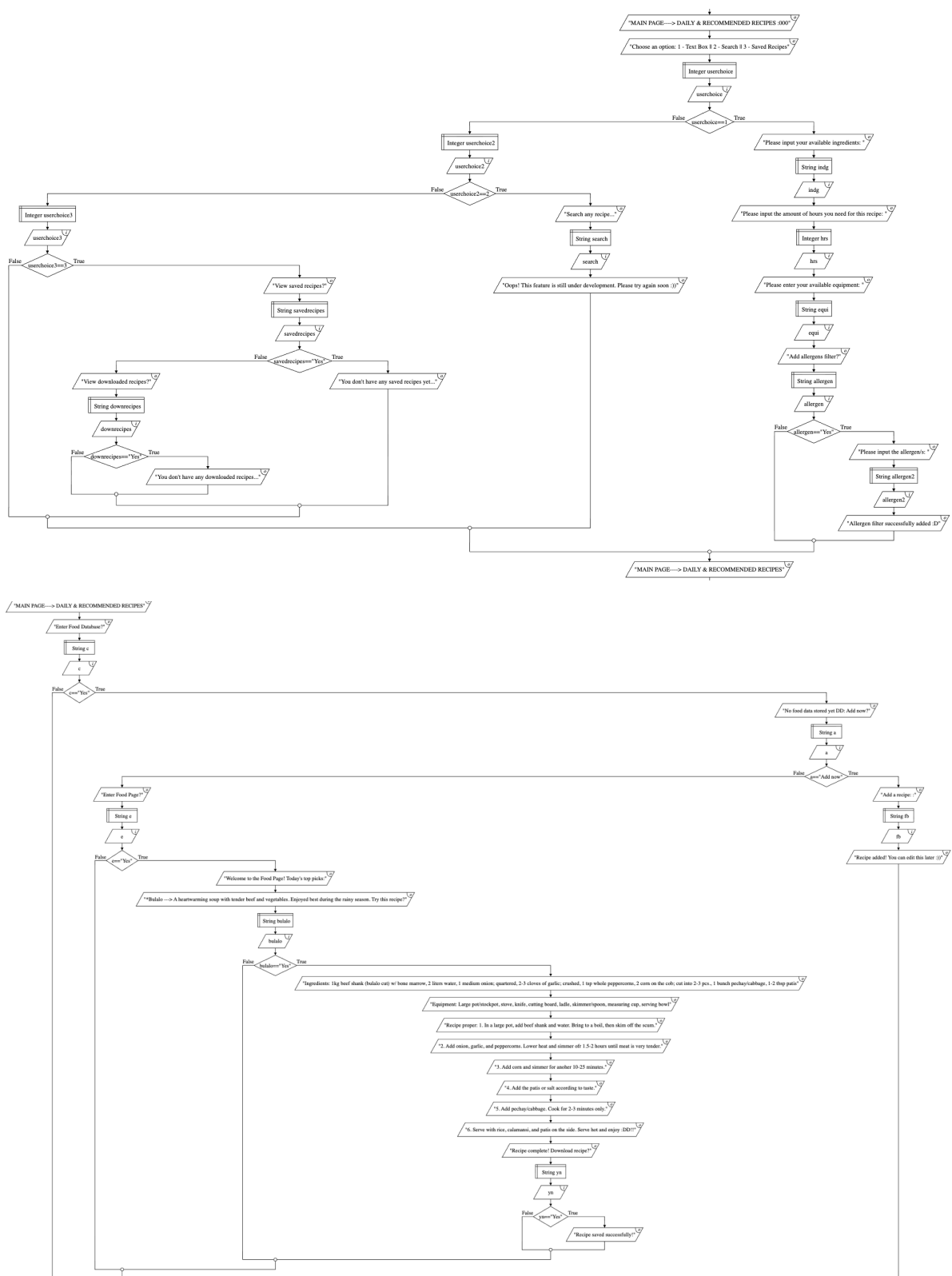
Recipes are shown depending on how many of the ingredients the user has. For every matched ingredient, the recipe gains one point. If an allergen is in the ingredient list of the recipe, the allergen state becomes true, warning the user.

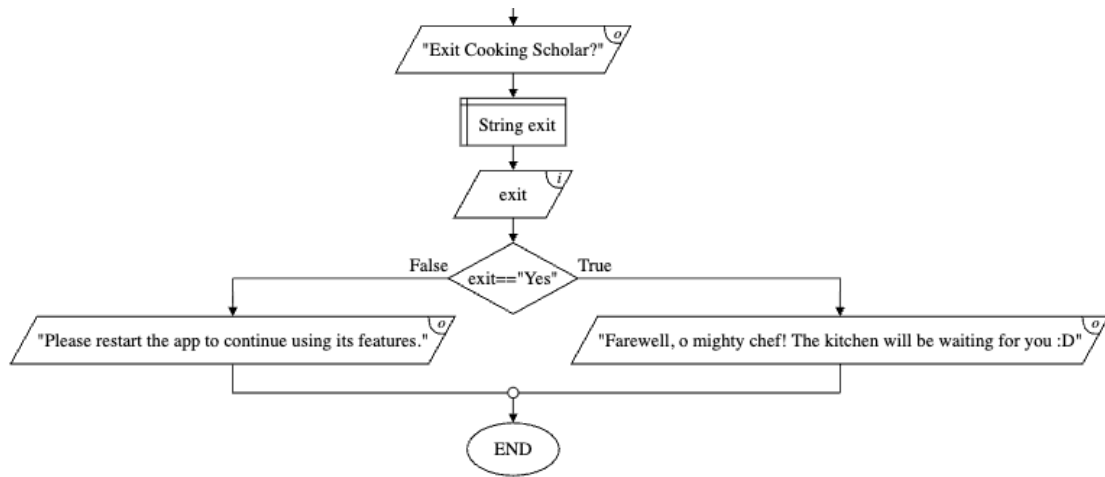
The project is now fully functional with more than 60% of features included. However, there are still essential features that are needed to make the project work effectively for launching and publicity. Features like storing user data so that their preferences are saved even after closing the program or 'main.py'.

Project Repository: <https://github.com/meterscenti/CookingScholar>

Flowchart of the Program (In parts)







CHAPTER 3 - USER END REQUIREMENTS

Minimum/recommended specifications to run the program:

- Processor: 2.0 GHz or higher
- RAM: 4.0 - 8.0 GB
- Storage: 50 GB HDD/SSD
- OS: Windows 10/11, Linux, or macOS

Programs needed to use Cooking Scholar:

- A Python IDE
- A file manager
- A terminal/command prompt

To install Cooking Scholar, go to the GitHub repository for the program, use git clone to copy the files and put them in a folder named CookingScholarv7, making sure that all the important files are copied. Don't rename any of the files, and make sure that all of the files have the same order as the ones in the repository. To run the program, click '**main.py**' to get started.

To add recipes in the database, go to files **ADELFA_Food_Database_COOKINGSCHOLAR** and **ADELFA_Food_Database_Search_COOKINGSCHOLAR**. In the first file, add the recipe's ID as the key and the instructions as the value in the dictionary. After adding the instructions in the first file, go to the second file, and find a dictionary named '**recipes**'. There, add the recipe name as the key and set the value to a list. In the list, the first item should have all of the ingredients needed for the recipe, the second item is the amount of ingredients it has in common with the user-input data, with default being 0, the third item is if it has an allergen, True if it has and False if not, and lastly, the fourth item is the recipe ID, the same ID in the first file with the instructions.

CHAPTER 4 - TESTING AND TROUBLESHOOTING

The testing phase for “Cooking Scholar” primarily focused on verifying its main functions to ensure a smooth user experience. The development of the application used a modular strategy, and testing was conducted throughout the process. The goals of this application included verifying recipe suggestions based on user inputs, such as ingredients, time, and equipment, as well as confirming the authenticity of the allergen filter and the interface's overall functionality. The testing involved the use of Python as the programming language, with tools like Python IDEs, file managers, and terminals, functioning on standard system arrangements.

The testing covered various situations to cover the application’s features. This process included the testing of ingredient-based recipe searches, with filters such as time and equipment limits, alongside the combined application fulfilling these requirements. There was more focus and effort committed to the allergen filter for the recipes with specified allergens to be properly excluded from said recipes. Additionally, scenarios were specially maneuvered to test the data management aspects, such as adding new recipes to the database files (ADELFA_Food_Database_COOKINGSCHOLAR & ADELFA_Food_Database_Search_COOKINGSCHOLAR)) and verifying that saved recipes were accessible regardless of online status. Other features of the application, such as the user interface, including the login, tutorial navigation, and the main page display, were also thoroughly evaluated to check for any bugs or errors.

Results from the testing state that “Cooking Scholar” has achieved over 95% functionality. The core features of ingredient matching and recipe scoring appear to be fully operational, and the allergen detection system is in place, allowing it to provide user warnings. The text-based UI enables the user to input essential information and fundamental functions such as the user login, search, and the save recipes function, which are well put together. However, the testing also indicated that approximately 4-5% of planned features have the problem of being the crucial functionality of saving user data for persistent preferences across sessions.

While the provided documentation does not specify the bugs encountered, it acknowledges the bugs and provides updates such as additions, changes, deletions, and bugs across different program versions (v1-v7). The troubleshooting involved code debugging, log analysis, and file integrity verification, particularly for the recipe database.

To conclude, the testing of “Cooking Scholar” has proved its foundational capabilities in recipe suggestion and allergen filtering, supported by a modular development form of approach. Key areas identified for further improvement include the implementation of user data and potentially enhancing the UI beyond its current text-based format to improve overall functionality and user experience. Further comprehensive testing, including performance and scalability assessments, was done to ensure the application’s durability.

CHAPTER 5 - CONCLUSION

The primary goal of the project was to provide an easily accessible mobile application to help students and busy individuals save time and resources by finding recipes using available ingredients and equipment. Following recipes from the mobile app rather than relying on outside food offers health and financial benefits.

The project is built entirely in Python, which is fully functional with over 60% of targeted features included. Necessary UI functions, namely, the Log-in page, which allows the user to log in to their account or sign up for one, the search bar for recipes, the food databases, the Allergen button, which applies filters to recipes to avoid allergens, and the Download button to save recipes, are all implemented through a text-based UI in the application.

Despite the many accomplishments of the app, there are still missing features that are vital for public use. Its limitations include the current text-based User Interface, which may serve as the foundation for future graphical, mobile-friendly button UI development. The remaining planned features are yet to be implemented. Additionally, for full accessibility, the app requires an internet connection. To further improve the app and qualify for publication, the limitations provided should act as the points of improvement for development.

Ultimately, Cooking Scholar has reached its objective to provide a healthy and cheaper alternative to outside food through the means of searching for recipes using the user inputs of available ingredients and equipment. Although missing features, it is a proven algorithmic foundation for mobile deployment, which can have a positive impact on the world, aiding preoccupied individuals around the globe.

Team roles

1. De La Pena, Don Emmanuel. - Lead Programmer
2. Banawan, Emmanuele - Interface Programmer
3. Ilao, Mc Brayden - Programmer and Documentation
4. Custodio, Zooey - Programmer and Idea Manager
5. Espiritu, Marriene - Programmer and Lead Tester

CODE (all in order):

```
Welcome to the Log-in Page!!!

    Please input your choice:
    [L] for Log-in;    [S] for Sign-up
    [INPUT]: S

    (If you chose the wrong option, type "0" in the username area!)
    Enter a new username: welove67
    Enter your chosen password: 0123one

    Welcome, welove67! Thank you for using Cooking Scholar!
```

```
Since you are new, have a quick tutorial

    -TEXT BOX-

    In the main page, the Text Boxes allows you to type in:
        --> the ingredients you have
        --> time available in minutes
    Press ENTER to continue.

    Example input:
        Bread, Eggs, Milk
        15
    Press ENTER to continue.

    -ALLERGEN BUTTON-

    If you have allergies, you can use the Allergen Button.
    This can filter out recipes with unwanted ingredients like:
        --> Milk
        --> Nuts
        --> Shrimp
    Press ENTER to continue.

    -SEARCH ENGINE-

    Looking for something?
    You can look for recipes with the Search Engine like:
        --> Pork Empanada
        --> Adobo
        --> Pancakes
    Press ENTER to continue.
```

```
    -FOOD PAGES-

    From our database, the Food Pages shows you the information of the recipe you searched for.
    Pages include:
        --> Ingredients
        --> Time to allot
        --> Instructions
    Press ENTER to continue.

    -DOWNLOAD BUTTON AND SAVED RECIPES-

    Found a recipe you absolutely love?
    Press the Download Button to save it and access it even offline in the Saved Recipes!
    Press ENTER to continue.

    And that's the tutorial!

    It's time for you to start cooking. Have fun!
```

```

Per aspera ad astra! Welcome to the Cooking Scholar (*~*)!

*RECIPES TO MAKE YOUR DAY*

    o Tinola
    o Adobo
    o Sinigang

Would you like to select a recipe, type 'no' if none? Sinigang

1. Rinse pork ribs and drain well.
2. In a pot over medium heat, combine pork and enough water to cover. Bring to a boil, skimming scum that accumulates on top.
3. Once broth clears, add tomatoes, onion, and fish sauce. Lower heat and simmer for about 1 to 1 ½ hours or until meat is tender, adding more water as necessary to maintain about 8 cups.
4. Add gabi and cook for about 4 to 6 minutes or until tender.
5. Add chili peppers and radish. Continue to simmer for about 2 to 3 minutes.
6. Add long beans. Continue to cook for about 2 minutes.
7. Add eggplant and okra and cook for another 1 to 2 minutes.
8. If using packaged tamarind base, add to the pot and stir until completely dissolved.
9. Season with salt and pepper to taste.
10. Add bok choy and continue to cook for about 1 minute. Serve hot.

```

```

*****COOKING SCHOLAR -- HOMEPAGE*****

HOMEPAGE INSTRUCTIONS:
Please enter
[1] To enter textbox,
[2] For your saved recipes,
[3] Exit.

1
Please Enter 1 ingredient, type 'done' to stop: fish
Please Enter 1 ingredient, type 'done' to stop: egg
Please Enter 1 ingredient, type 'done' to stop: flour
Please Enter 1 ingredient, type 'done' to stop: done
Please Enter 1 allergen that you have, type 'done' to stop: done

Fish Fillet: 100.0% matching ingredients, Contains Allergen: False, ID: 1
Egg Sandwich: 33.33% matching ingredients, Contains Allergen: False, ID: 3
Fried Rice: 0.0% matching ingredients, Contains Allergen: False, ID: 2
Tinola: 0.0% matching ingredients, Contains Allergen: False, ID: 4
Adobo: 0.0% matching ingredients, Contains Allergen: False, ID: 5
Sinigang: 0.0% matching ingredients, Contains Allergen: False, ID: 6
What recipe do you want to cook(ID):

```

```

===Fish Fillet (15 minutes)===
1. Cut the fish into strips.
2. Dip in egg mixture, you can add seasonings if you like.
3. Drop in some flour.
4. Deep fry in high heat until golden brown.

Would you like to save this recipe(Yes/No): Yes

```

```

HOMEPAGE INSTRUCTIONS:
Please enter
[1] To enter textbox,
[2] For your saved recipes,
[3] Exit.

2
***SAVED RECIPES***

===Fish Fillet (15 minutes)===
1. Cut the fish into strips.
2. Dip in egg mixture, you can add seasonings if you like.
3. Drop in some flour.
4. Deep fry in high heat until golden brown.

```

```

*****COOKING SCHOLAR -- HOMEPAGE*****

HOMEPAGE INSTRUCTIONS:
Please enter
[1] To enter textbox,
[2] For your saved recipes,
[3] Exit.

3
Closing Program...
>>>

```

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